

Kathan Joshi

Clarksburg, MD | 240-390-5282 | joshikathan03@gmail.com | [linkedin.com/in/kjoshi03/](https://www.linkedin.com/in/kjoshi03/) | [Portfolio](#)

EDUCATION

Pennsylvania State University (3.67/4.00)

Master of Science in Computer Science

Pennsylvania, USA

Aug 2023 – May 2025

Ahmedabad University (3.41/4.00)

Bachelor of Technology in Computer Science and Engineering

Gujarat, India

Aug 2019 – Jun 2023

WORK EXPERIENCE

AI Product Owner and Lead AI Engineer (Volunteer)

Aug 2025 – Present

WinWin Labs

Remote, USA

- Designed and implemented agentic AI architectures leveraging LLMs (GPT-4, Llama) with retrieval-augmented generation (RAG), tool/function calling and fallback policies; improved recall@10 by 23% and answer accuracy by 20%.
- Developed APIs and microservices to expose AI capabilities, integrating with cloud platforms (AWS/GCP), vector databases and message queues; decreased inference latency 28% through caching and parallel processing.
- Created evaluation harnesses and A/B testing frameworks to measure faithfulness, recall@k, MRR and answer similarity; reduced hallucination rate 15% and improved overall response quality by 12 pp.
- Partnered with cross-functional teams to translate product requirements into technical specifications, implement PII redaction and privacy-by-design, and deliver features that met compliance and user needs.

Machine Learning Research Assistant/TA

Aug 2023 – May 2025

Pennsylvania State University

Pennsylvania, USA

- Engineered and fine-tuned a suite of 7 ML models (AdaBoost, CatBoost, XGBoost, Random Forest, KAN, MLR, NN), to predict the LOS of a hospital patient, improving performance by 12%.
- Performed EDA on a 60 M-record healthcare dataset using Python, SQL, and Dask; collaborated with clinical researchers to surface LOS drivers and delivered Tableau dashboards — improving predictive performance by 25%.
- Orchestrated production-ready data pipelines (Python, Dask) for feature engineering; integrated automated data quality checks and MLflow model tracking—ensuring reproducibility and reducing inconsistencies by 30%.
- Developed model interpretability reports using SHAP to validate fairness and support transparent decision-making, aligning with ethical AI guidelines.

Data Scientist

Aug 2022 – Jul 2023

Safal Industries

Gujarat, India

- Performed EDA on equipment and production data using Python, Pandas, and AWS SageMaker, identifying trends that increased productivity by 11% and reduced costs by 7%, presented to stakeholders for strategic decisions.
- Engineered an LSTM time-series forecasting model for 5 SKUs, reducing WAPE from 11% to 7%; deployed via Docker on AWS SageMaker with auto-scaling pipelines to support production load.
- Implemented real-time anomaly detection using Spark Streaming + XGBoost in Docker/Kubernetes, accelerating quality-control by 20%.
- Fine-tuned a BERT transformer for sentiment analysis on 6K reviews; integrated outputs into stakeholder dashboards and influenced product changes to reduce churn by 8%.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, Java, JavaScript, HTML/CSS, C/C++

Data Wrangling & Math: NumPy, Pandas, Dask, SciPy, Statsmodels

ML/DL: scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, PyTorch, Spacy

Time-Series & MLOps: Prophet, Darts, MLflow, Docker, K8s, AWS, GCP, Spark, Git

NLP/GenAI & BI: Transformers, LangChain, LangGraph, CrewAI, LlamaIndex, FAISS, BERT, AutoGen | Tableau, PowerBI

Certifications: [Salesforce Agentforce Specialist](#), [AWS AI Certified](#), [Machine Learning Specialization](#), [GenAI with LLMs](#)

PROJECTS

ResumeSense: AI-Agentic ATS | *SpaCy, Transformers, LangChain, FAISS, Flask, Docker*

Mar 2025 – Present

- Architected a multi-agent system for NER, similarity scoring, summarization, and question generation; designed structured system/user prompts, boosting throughput by 55% and cutting manual workload by 70%.
- Built a RAG chatbot (LangChain + FAISS) with guardrails and prompt templates; reduced response latency by 40% and improving response accuracy by 23%; containerized and deployed on Hugging Face Spaces.

KATNet | *Pytorch, NumPy, Pandas, SKlearn, Htmld, Moleculekut, Matplotlib*

Jan 2024 – Aug 2024

- Engineered a novel 3D-CNN with an integrated cross-attention mechanism in PyTorch for protein-ligand binding affinity prediction, accelerating training convergence by 25%.
- Outperformed baseline models on a PDBbind (v.2016) subset by boosting R-squared by 6% and reducing RMSE by 1.37, underscoring superior predictive performance.